

Average Marginal Effects (AME) Implementation in Sleepiness Pipeline

The AME implementation in the sleepiness pipeline (specifically within `estimation_5_sleep.py`'s `get_nlls_ame_and_se` function) uses a **hybrid approach: Analytical for the effects, and Numerical for the Standard Errors (via Delta Method)**.

Here is exactly what it is computing under the hood:

1. Computing the Marginal Effects (Analytical)

The script iterates through every covariate in your non-linear least squares estimation and checks if it's a binary dummy (only 0s and 1s) or a continuous variable. It then applies the appropriate analytical formula based on the Logit CDF:

- **For Dummy Variables:**

It computationally identical twins your entire dataset—in one copy, it forces the dummy variable to 1, and in the other, it forces it to 0. It passes both matrices through the predicted probability function and takes the mean of the exact differences.

Analytical formula: $AME = \text{mean}(P_1 - P_0)$

- **For Continuous Variables:**

It computes the exact analytical derivative of the logistic function for each observation, multiplied by the estimated coefficient θ_k , and takes the average.

Analytical formula: $AME = \text{mean}(P * (1 - P) * \text{theta}_k)$

2. Computing the Adjusted Standard Errors (Numerical)

Because marginal effects are non-linear transformations of your estimated parameters $\hat{\theta}$, the Standard Errors must be appropriately scaled using the Delta Method: $Cov_{AME} \approx J \cdot Cov(\hat{\theta}) \cdot J^T$.

To get the Jacobian matrix (J) required for the Delta Method, the script uses a **Numerical Derivative**:

- It loops through every estimated parameter and perturbs it by a tiny finite-difference step size ($h = 1e-5$).

- It recalculates the analytical AMEs under the perturbed parameter (`ame_step`).
- It populates the Jacobian simply as $(\text{ame_step} - \text{AME}) / h$.

So, in short: the **effects themselves are exact analytical derivatives** (or discrete differences for dummies), but the **Jacobian used to adjust the Standard Errors is taken numerically**.